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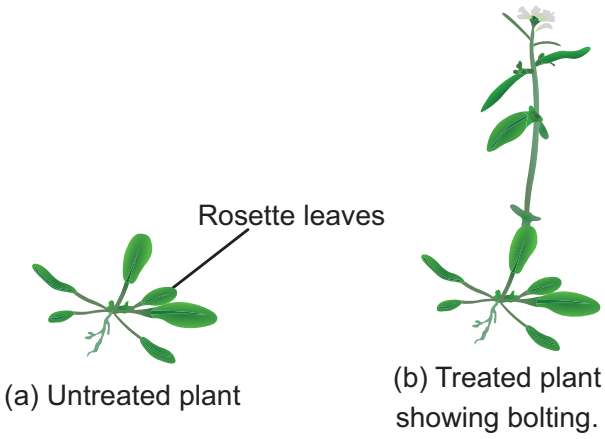


Figure 15.13: Bolting

exposure to cold season in the first year itself, when they are treated with gibberellins.

8. Agricultural role

- Formation of seedless fruits without fertilization is induced by gibberellins Example: Seedless tomato, apple and cucumber.
- It promotes the formation of male flowers in cuccurbitaceae. It helps in crop improvement.
- Uniform bolting and increased uniform seed production.
- Improves number and size of fruits in grapes. It increase yield.
- Promotes elongation of inter-node in sugarcane without decreasing sugar content.
- Promotion of flowering in long day plants even under short day conditions.
- It stimulates the seed germination.

15.2.3 Cytokinins (*Cytos* – cell, *Kinesis* – division)

1. Discovery

The presence of cell division inducing substances in plants was first demonstrated by **Haberlandt** in 1913 in Coconut milk (liquid endosperm of coconut) which contains cell division inducing substances. In 1954, Skoog and Miller discovered that autoclaved DNA from herring sperm stimulated cell division in tobacco pith cells. They called this cell division inducing principle as kinetin (chemical structure: 6-Furfuryl Amino Acid). This does not occur in plants. In 1963, Lethan introduced the term cytokinin. In 1964, Lethan and Miller isolated and identified a new cytokinin called **Zeatin** from unripe grains of maize. The most widely occurring cytokinin in plants is Iso Pentenyl adenine (IPA).

2. Occurrence

Cytokinin is formed in root apex, shoot apex, buds and young fruits.

3. Precursor

Cytokinins are derivatives of the purine adenine.

4. Bioassay (Neem Cotyledon Assay)

Neem cotyledons are measured and placed in cytokinin solution as well as in ordinary water. Enlargement of cotyledons is an indication of cytokinin activity.

5. Transport in plants

The distribution of cytokinin in plants is not as wide as those of auxin and gibberellins but found mostly in roots. Cytokinins appear to be translocated through xylem.

6. Physiological effect

- Cytokinin promotes cell division in the presence of auxin (IAA).
- Induces cell enlargement associated with IAA and gibberellins